SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Short Answer Test

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0512

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.* (BTL 3)

Assessment Tool Weightage: 40%

QUESTIONS

1. A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.
2. Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.
3. An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.
4. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.
5. A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.
6. Explain how image formation models help in improving image quality in applications such as autonomous driving.
7. A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.
8. Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.
9. In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.
10. An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.
11. Compare two image acquisition systems (CCD vs CMOS) in terms of their impact on image quality in practical applications.
12. A digital image appears distorted due to improper image formation. Identify possible causes and suggest improvements based on image formation models.
13. Explain how the features of Computer Vision systems enable applications such as automated surveillance.
14. A system processes satellite images for land analysis. Explain how digital image fundamentals and acquisition methods influence the accuracy of results.
15. Given an application like OCR (Optical Character Recognition), explain how sampling, quantization, and image acquisition collectively affect performance.

16. A high-resolution image consumes excessive storage and processing time. Explain how sampling and quantization can be adjusted to optimize performance without significant quality loss.
17. An image captured under uneven lighting shows poor contrast. Explain how the image formation process contributes to this issue and suggest corrective measures.
18. In a medical imaging system, explain how improper quantization can lead to loss of critical diagnostic information.
19. A self-driving car system relies on visual input. Explain how image acquisition and Computer Vision features contribute to reliable decision making.
20. An application requires detecting fine details in images. Explain how sampling rate and intensity resolution should be chosen to meet this requirement.
21. An image captured at a low sampling rate shows jagged edges. Explain how sampling rate affects spatial resolution and suggest improvements.
22. A digital camera produces images with banding effects. Analyze how quantization levels contribute to this issue and propose a solution.
23. In a thermal imaging system, explain how image sensing affects temperature-based visualization accuracy.
24. A blurred image is obtained due to motion during acquisition. Explain how image formation and sensing contribute to this effect.
25. A satellite image shows loss of fine terrain details. Explain how sampling and resolution affect feature visibility.
26. In a biometric system, explain how image acquisition quality affects recognition accuracy.
27. An image appears pixelated when zoomed. Analyze the role of sampling and resolution in this effect.
28. A digital image shows reduced contrast due to poor sensing conditions. Explain how acquisition parameters influence this.
29. In a machine vision system, explain how improper quantization affects defect detection.
30. A camera captures images with varying brightness levels. Explain how image formation models influence intensity variations.
31. An image processing system fails to detect edges clearly. Explain how sampling and acquisition affect edge visibility.
32. In an industrial inspection system, explain how image sensing impacts defect detection reliability.
33. A low-bit image loses important details. Explain how quantization affects information preservation.
34. A surveillance image appears noisy in low light. Analyze how sensing and acquisition conditions contribute to noise.
35. A digital image suffers from aliasing when resized. Explain how sampling principles apply in this scenario.
36. In medical imaging, explain how increasing sampling rate improves diagnostic accuracy.
37. A system captures images with inconsistent quality. Explain how acquisition methods influence uniformity.

38. A high-speed camera captures distorted images. Explain how image formation and sampling contribute to distortion.
39. In an OCR system, explain how poor sampling affects character recognition accuracy.
40. A digital image shows quantization noise. Explain its origin and suggest mitigation techniques.
41. A vision system processes images with varying illumination. Explain how image formation models help normalize such variations.
42. An image appears overly smooth with loss of detail. Explain how quantization levels contribute to this issue.
43. In a facial recognition system, explain how resolution impacts feature extraction accuracy.
44. A camera system produces images with color distortion. Explain how sensing and acquisition affect color representation.
45. A digital image has insufficient contrast. Explain how image acquisition and formation influence contrast levels.
46. A system requires detecting small objects in images. Explain how sampling and resolution should be adjusted.
47. In satellite imaging, explain how quantization affects classification accuracy.
48. A vision system fails in low-light conditions. Explain how image sensing impacts system performance.
49. A digital image contains staircase artifacts. Explain how sampling causes this effect.
50. In an industrial system, explain how acquisition noise affects measurement accuracy.
51. A camera captures images with motion blur. Explain how image formation and acquisition timing affect clarity.
52. A digital image has poor dynamic range. Explain how quantization influences intensity representation.
53. In a real-time vision system, explain how acquisition speed impacts processing efficiency.
54. A system captures images with uneven resolution across regions. Explain how sampling contributes to this issue.
55. In medical imaging, explain how image acquisition affects diagnostic reliability.
56. A digital image appears overly compressed. Explain how sampling and quantization contribute to information loss.
57. A vision system detects objects incorrectly due to poor acquisition. Explain the relationship between sensing and detection accuracy.
58. A high-resolution image still lacks detail. Explain how image formation models influence clarity.
59. In an automated inspection system, explain how sampling affects defect detection sensitivity.
60. A camera captures images with inconsistent intensity levels. Explain how quantization and sensing contribute to this variation.

Code of Conduct:

I **A.N.L.NARASIMHA(192525267)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Assessment Tool 1 – Short Answer Test: Answer Key

Course: Computer Vision (ITA05) | CO1 | Weightage: 40%

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1. Blur from improper sampling

Sampling decides how many pixels represent the scene. Too few samples lose fine detail and cause blur or aliasing. Correct it by increasing the sampling rate (higher resolution) or applying an anti-aliasing low-pass filter before sampling, following the Nyquist criterion (sample at ≥ 2× the highest frequency).

2. Low-light sensing and acquisition

In low light, few photons reach the sensor, so the signal is weak compared with sensor noise. The image comes out dark, noisy and low in contrast. Longer exposure, wider aperture, higher ISO or added illumination improves it.

3. Aliasing artifacts

Aliasing is caused by undersampling: the sampling rate is below the Nyquist rate, so high frequencies appear as false patterns (jaggies, moiré). Coarse quantization worsens the visual effect. Fix with anti-aliasing filtering before sampling or a higher sampling rate.

4. Pixel vs intensity resolution

Pixel (spatial) resolution is the number of pixels, which controls how small a structure can be seen, such as a tiny lesion. Intensity resolution is the number of gray levels (bits), which controls how subtle tissue-contrast differences appear. Medical imaging needs both high.

5. Level of CV for real-time surveillance

It is mid-level to high-level vision. Low-level handles filtering and enhancement, mid-level does segmentation and tracking, and high-level interprets behavior or recognizes objects for decisions. Real-time surveillance must detect, track and interpret events, so high-level is involved.

6. Image formation models in autonomous driving

They describe how scene light, geometry (camera model) and sensor response produce the image. Knowing this lets the system correct lens distortion, illumination changes and exposure, and estimate depth. The result is cleaner, more reliable input for detection.

7. Quantization of a grayscale image

Quantization maps continuous intensity values to a finite number of gray levels (e.g., 256 for 8 bits). Fewer levels reduce storage but cause false contours and loss of subtle detail. More levels preserve smooth tonal variation.

8. Noise from acquisition

Noise comes from photon shot noise, thermal/dark current, readout and electronic circuitry, especially at high ISO or temperature. Minimize it by cooling the sensor, using adequate lighting, lower ISO, longer exposure, frame averaging and denoising filters.

9. Levels of CV in facial recognition

Acquisition captures the face image. Low-level processing enhances and normalizes it (noise removal, illumination correction). Mid-level detects and segments the face and extracts features. High-level matches features with a database and decides identity.

10. Low resolution and object visibility

Low sampling gives too few pixels per object, so edges and details merge and small objects vanish. Poor image formation (blur from lens, distance, low light) lowers contrast further. Higher resolution, better optics and closer or zoomed capture improve visibility.

11. CCD vs CMOS

CCD gives high image quality with low noise and good uniformity, but uses more power and costs more, suiting scientific and astronomy imaging. CMOS is cheaper, faster, low-power with on-chip processing, and modern versions have comparable quality, suiting phones and real-time systems. CMOS has historically had slightly more noise.

12. Distortion from improper image formation

Causes include lens aberrations (barrel or pincushion), wrong focus, bad camera angle/perspective, uneven lighting and sensor misalignment. Improve by camera calibration, distortion-correction models, proper focus and lighting, and better lenses.

13. CV features enabling automated surveillance

Key features such as object detection, tracking, motion analysis, recognition and real-time processing let the system monitor scenes continuously. They automatically detect intrusions, track people or vehicles and raise alerts without human watching.

14. Satellite images for land analysis

Spatial resolution decides the smallest land feature seen, and quantization/radiometric resolution decides how well subtle differences (crop health, water) are distinguished. Spectral bands, sensor noise, atmospheric effects and acquisition angle also affect classification accuracy.

15. OCR performance

Adequate sampling keeps character strokes clear, enough gray levels keep text-background contrast, and good acquisition (lighting, focus, no skew) avoids noise and blur. Poor values in any of these cause broken or merged characters and misrecognition.

16. Optimizing a high-resolution image

Reduce sampling by downsampling (with anti-aliasing) to the resolution the task really needs. Reduce quantization levels (e.g., 8 to 6 bits) where fine tone isn't critical. Region-of-interest processing and compression also cut cost with little visible loss.

17. Uneven lighting and poor contrast

In image formation, image intensity = illumination × reflectance. Uneven illumination makes the same surface appear with different brightness, reducing contrast. Correct with uniform or controlled lighting, histogram equalization, homomorphic filtering or background normalization.

18. Improper quantization in medical imaging

Too few gray levels merge close intensities, so subtle differences between tissues, small lesions or early tumors disappear, and false contours appear. This can hide critical diagnostic information, so higher bit depth (12–16 bit) is used.

19. Self-driving car: acquisition and CV features

Cameras provide high-quality, timely images, and CV features such as object detection, lane detection, depth estimation and tracking interpret them. Together, and often fused with LiDAR/radar, they supply accurate, real-time information for safe decisions.

20. Fine-detail detection

Choose a high sampling rate so pixels are smaller than the finest detail (obey Nyquist, at least 2 samples per detail). Choose high intensity resolution (more bits) to capture small contrast differences. Balance against storage and speed.

21. Jagged edges at low sampling rate

A low sampling rate gives lower spatial resolution, so diagonal and curved edges are represented by coarse pixel steps (jaggies). Improve by increasing sampling rate/resolution, anti-aliasing, or interpolation/smoothing.

22. Banding effects

Banding (false contouring) arises when too few quantization levels represent smooth gradients, so visible steps appear. Use more bits per pixel (e.g., 10–12 bit) or dithering to hide the steps.

23. Thermal imaging sensing

Sensor sensitivity, calibration, noise, resolution and emissivity handling determine how accurately infrared radiation maps to temperature. Poor sensing or ambient/atmospheric effects give wrong temperature readings and blurred thermal detail.

24. Motion blur during acquisition

In image formation, the sensor integrates light over the exposure time. If the object or camera moves during that time, light from many positions spreads over multiple pixels, causing blur. Use shorter exposure, faster shutter or stabilization.

25. Satellite: loss of terrain details

Coarse sampling (large ground sample distance) means each pixel covers a big area and averages out small features. Lower spatial resolution therefore hides fine terrain features. Use higher-resolution sensors or finer sampling.

26. Biometric acquisition quality

Good acquisition (sharp focus, proper lighting, adequate resolution, low noise) captures reliable features such as fingerprints, iris or face. Poor quality causes missing or wrong features, raising false rejects and false accepts.

27. Pixelation on zoom

The image has limited spatial resolution (few samples), so zooming enlarges individual pixels into visible blocks. Missing detail cannot be recovered; interpolation (bilinear, bicubic) only smooths it. Capture at higher resolution.

28. Reduced contrast from poor sensing

Parameters such as exposure time, aperture, ISO/gain, sensor dynamic range and lighting control signal range. Under- or over-exposure compresses intensities into a narrow range, lowering contrast. Adjust them or apply contrast stretching.

29. Machine vision: improper quantization

Too few levels hide small intensity changes, so faint scratches, cracks or stains merge with the background and defects go undetected, or false contours cause false alarms. Use sufficient bit depth.

30. Brightness variation and formation models

Brightness depends on $f(x,y) = i(x,y) \times r(x,y)$: illumination times reflectance. Changes in light source, angle or surface reflectance change intensity values. The model explains this and allows illumination to be separated or corrected.

31. Edges not detected clearly

Low sampling blurs and merges edges, and noise, blur or poor lighting during acquisition reduce edge contrast. Use higher resolution, proper focus, better lighting and noise reduction before edge detection.

32. Industrial inspection: sensing

The sensor's resolution, sensitivity, noise level and speed determine whether small defects are captured clearly and consistently. Poor sensing misses defects or gives false detections, reducing reliability.

33. Low-bit image losing details

With few bits, intensities are grouped into few gray levels, so gradual variations and fine textures are lost and false contours appear. Information preservation improves as bits per pixel increase.

34. Noisy surveillance image in low light

Low light means fewer photons, so a low signal-to-noise ratio. The camera raises gain/ISO, which amplifies sensor and electronic noise. Add illumination, use larger sensors or longer exposure, and apply denoising.

35. Aliasing when resizing

Downsampling reduces the sampling rate below Nyquist for the image's high frequencies, producing aliasing. Apply a low-pass (Gaussian) filter before downsampling, or use area/bicubic resampling.

36. Higher sampling rate in medical imaging

More samples give finer spatial resolution, so small structures like micro-calcifications or tiny lesions become visible. This reduces missed findings and improves diagnostic accuracy, at the cost of storage.

37. Inconsistent image quality

Variation in lighting, camera settings, sensor type, focus, distance and noise across captures causes non-uniform quality. Standardized acquisition (fixed lighting, calibrated settings, same sensor) gives uniform images.

38. High-speed camera distortion

Very short exposure gives little light (noise), and sequential (rolling shutter) readout captures rows at different times, distorting fast objects. Insufficient frame/sampling rate also causes temporal aliasing. Use global shutter and proper lighting.

39. Poor sampling in OCR

Low sampling makes character strokes thin, broken or merged, so similar characters (e.g., 'o' and 'c') are confused. Recognition accuracy drops. Scan at adequate resolution (typically ≥ 300 dpi).

40. Quantization noise

It originates from rounding continuous intensity values to the nearest discrete level, giving error up to half a step. Reduce it with more quantization levels (higher bit depth), dithering or noise-shaping.

41. Varying illumination and formation models

Because intensity = illumination × reflectance, models help estimate and remove the illumination component (e.g., homomorphic filtering, retinex). This normalizes brightness so the reflectance (object properties) remains.

42. Overly smooth image with detail loss

Very few quantization levels (and smoothing) remove small intensity differences, so textures and subtle edges are flattened. Increasing the number of levels restores tonal detail.

43. Resolution in facial recognition

Higher resolution gives more pixels on the face, so features such as eyes, nose and jawline can be extracted accurately. Low resolution blurs landmarks and reduces matching accuracy.

44. Color distortion

Sensor color filter array (Bayer), spectral response, white balance, illumination color and ADC accuracy determine color. Errors in them shift colors. Fix with white balance, color calibration and proper lighting.

45. Insufficient contrast

Acquisition factors (exposure, aperture, gain, sensor range) and formation factors (lighting, reflectance) limit intensity spread. Correct lighting and exposure, or use histogram equalization/stretching.

46. Detecting small objects

Increase sampling/resolution so small objects span enough pixels (a few pixels at minimum), use a closer or zoom lens and a higher-resolution sensor. Avoid excessive downsampling.

47. Quantization in satellite classification

More bits (radiometric resolution) separate similar land-cover reflectances (e.g., different crops). Coarse quantization merges classes and lowers classification accuracy.

48. Low-light failure

Weak signal and high noise from the sensor make features hard to extract, so detection and recognition fail. Better sensitivity (larger pixels, high quantum efficiency), IR/illumination and denoising improve performance.

49. Staircase artifacts

Low sampling approximates slanted or curved edges by square pixels, producing step-like jaggies. Higher sampling rate or anti-aliasing reduces them.

50. Acquisition noise in industrial measurement

Noise randomly shifts pixel intensities and edge positions, causing errors in dimension and position measurements and reducing repeatability. Reduce with stable lighting, averaging and good sensors.

51. Motion blur and timing

Image formation integrates light during exposure; motion during this time smears the image. Short exposure/fast shutter, strobe lighting or synchronizing capture with motion keeps images sharp.

52. Poor dynamic range

Few quantization levels or limited bit depth cannot represent both dark and bright regions, so shadows or highlights clip or flatten. Higher bit depth/HDR extends the representable intensity range.

53. Acquisition speed in real-time systems

Higher frame rates give more up-to-date data but increase data volume, so processing must keep up. If acquisition is slower than needed, motion is missed; if too fast, processing lags. Matching speeds ensures smooth real-time operation.

54. Uneven resolution across regions

Non-uniform sampling grids, lens distortion, perspective or varied object distance mean some regions get fewer samples per unit area. Uniform sampling and distortion correction fix this.

55. Acquisition in medical imaging

Resolution, noise, contrast and correct exposure/calibration determine how clearly anatomy appears. Poor acquisition leads to artifacts or missed findings, lowering diagnostic reliability.

56. Overly compressed image

Reduced sampling (downsizing) discards spatial detail, and coarser quantization removes tonal detail. Together they cause blockiness, blur and lost information. Use higher resolution/bit depth or lossless compression.

57. Sensing and detection accuracy

Good sensing (sensitivity, resolution, low noise) gives clear object features, so detection is accurate. Poor sensing gives noisy, blurred or low-contrast data, leading to missed or wrong detections.

58. High resolution but lacking detail

Image formation factors like poor focus, lens blur, diffraction, motion, haze or low contrast reduce clarity regardless of pixel count. Formation models (such as point spread function) help identify and deblur or correct these.

59. Sampling and defect detection sensitivity

A finer sampling grid lets smaller defects cover multiple pixels and be detected; coarse sampling can miss or blur them. Defect size should span several pixels.

60. Inconsistent intensity levels

Variations in sensor gain, noise, exposure and illumination change the measured signal, and coarse or unstable quantization maps similar signals to different levels. Calibrate the sensor, stabilize lighting and use adequate bit depth.